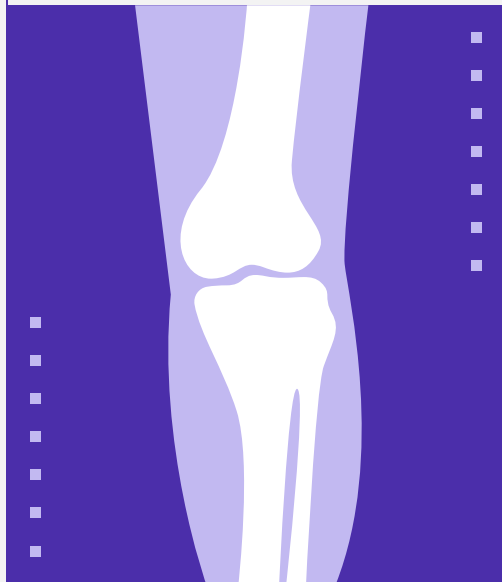




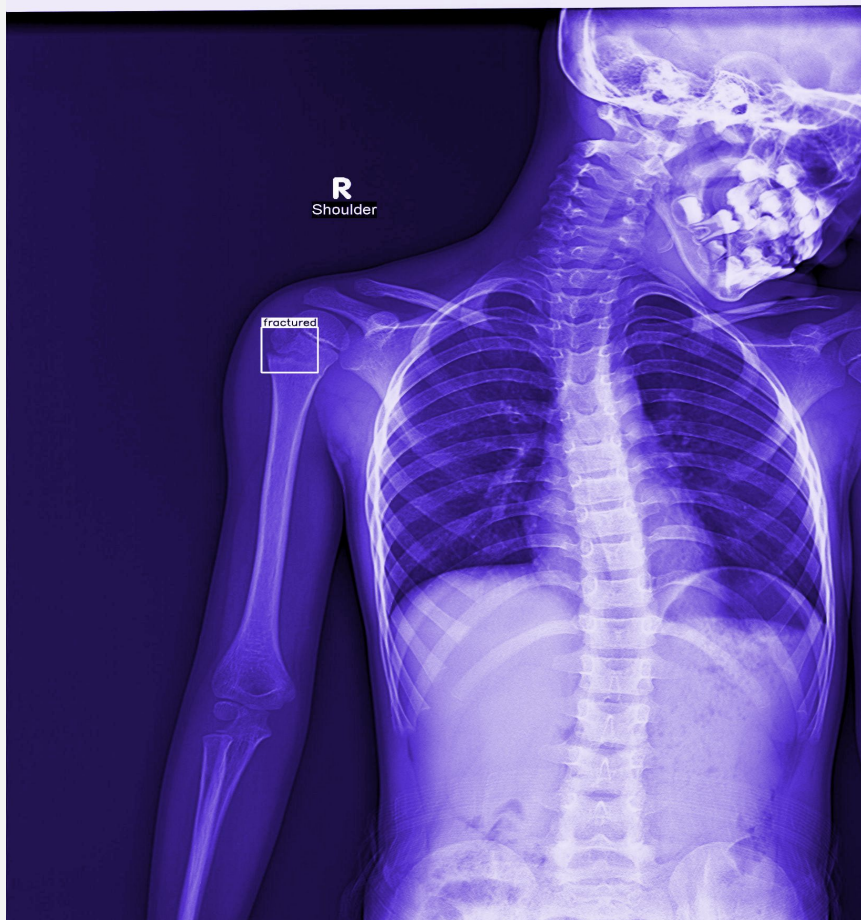
BONE FRACTURE LOCALIZATION USING YOLO

Mimmy Mei, Sophia Huang, Charlotte Li, McKenzie Skrastins



01

BACKGROUND



BACKGROUND

AI detection on medical images

Medical diagnosis

Common injuries

Diagnosis accuracy





02

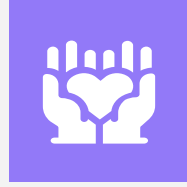
OUR GOALS



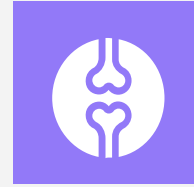
OUR GOALS



EFFICIENCY



PATIENT
OUTCOMES



RELIEVE
PRESSURE

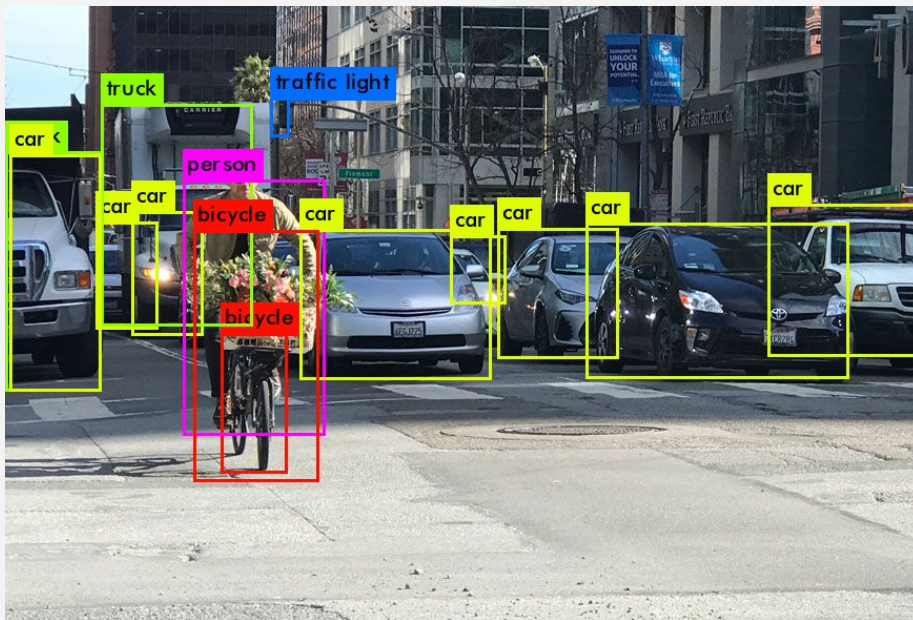




03

MODEL ARCHITECTURE

You Only Look Once (YOLO)



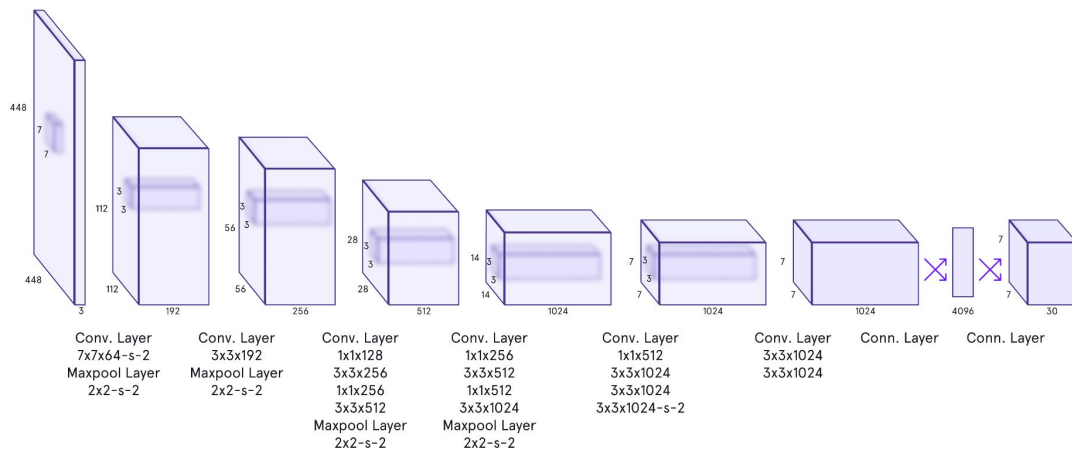
Object Detection

Multiple Class Identification

"One Pass"



You Only Look Once (YOLO)

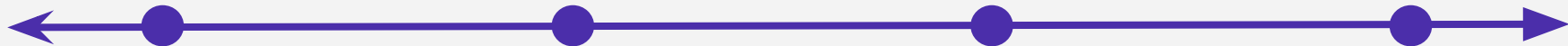


+ **post processing**

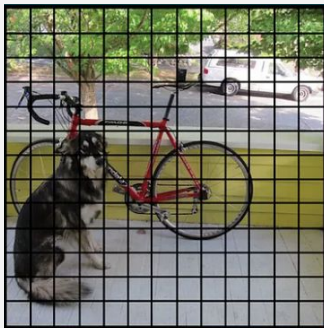




YOLO

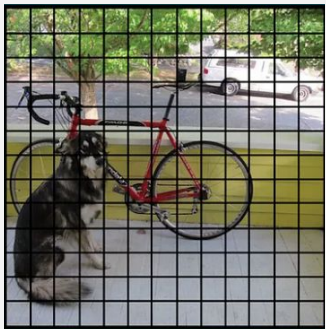


**Making
Grid**



YOLO

Making
Grid

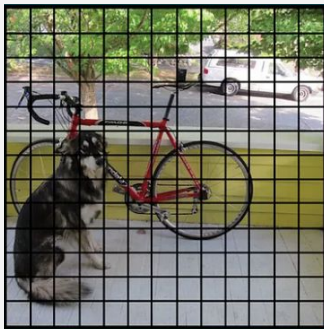


Generating
Bounding Boxes



YOLO

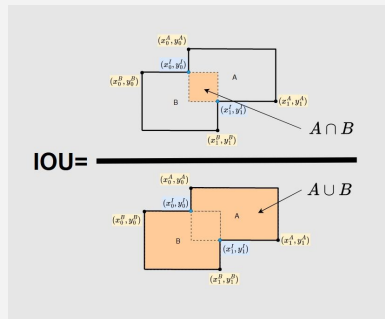
Making
Grid



Generating
Bounding Boxes

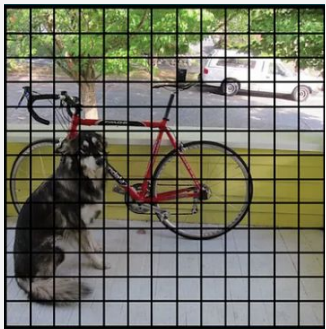


**Non-Max
Suppression**



YOLO

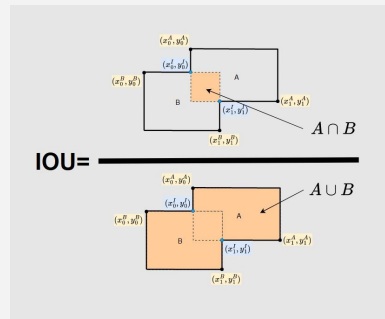
Making
Grid



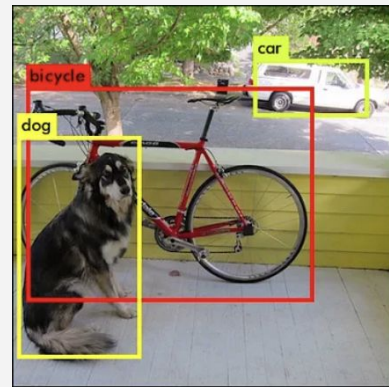
Generating
Bounding Boxes



Non-Max
Suppression



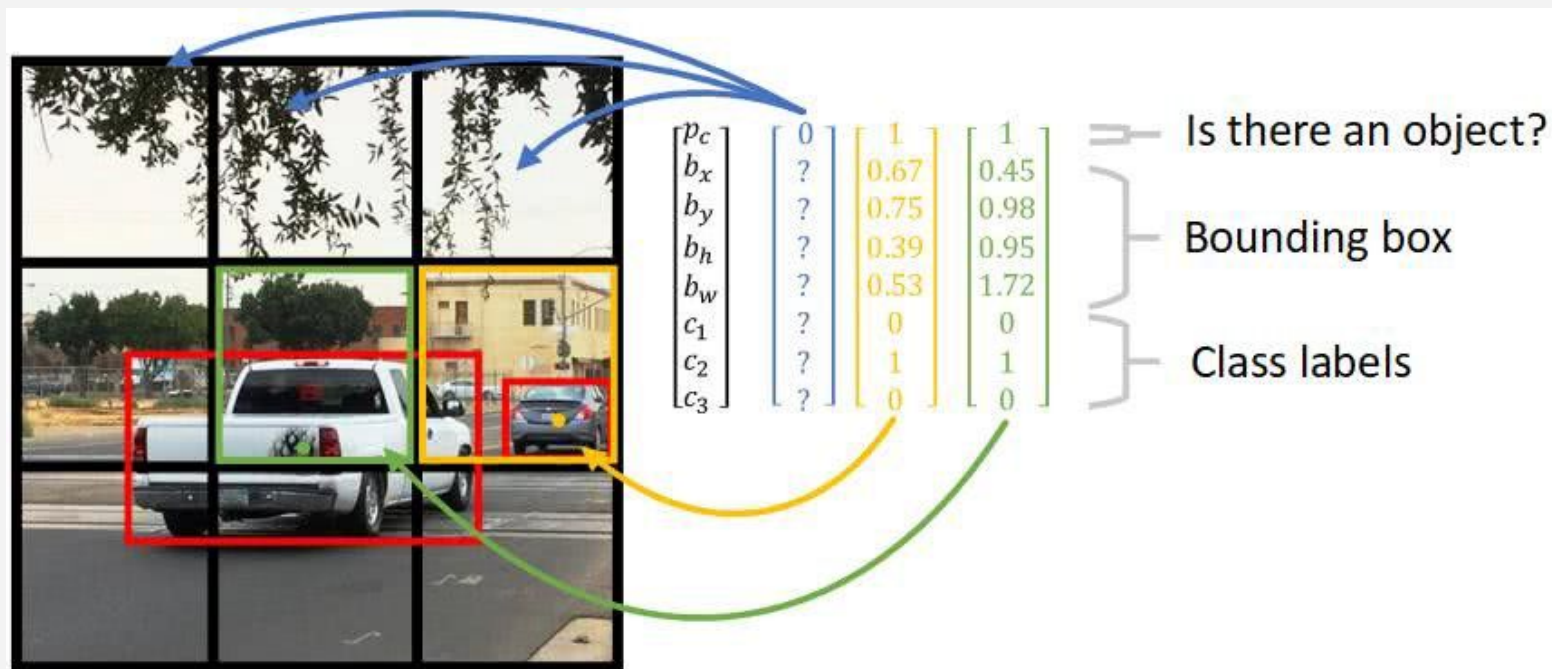
**Select Best
Box**



YOLO: Making Grid

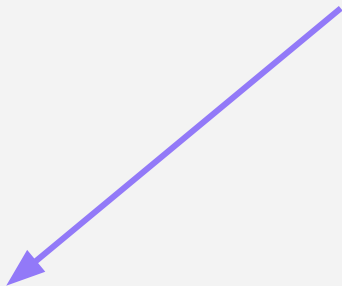


YOLO: Generating Bounding Boxes



YOLO: Generating Bounding Boxes

$$Y = [pc, bx, by, bh, bw, c1]$$



**Probability the
box contains
an object**

****p.b.b = predicted bounding box****

YOLO: Generating Bounding Boxes

$$Y = [pc, bx, by, bh, bw, c1]$$



Probability the
box contains
an object

**X coordinate
of p.b.b.**

****p.b.b = predicted bounding box****

YOLO: Generating Bounding Boxes

$$Y = [pc, bx, by, bh, bw, c1]$$

Probability the
box contains
an object

X coordinate
of p.b.b.

**Y coordinate
of p.b.b.**

****p.b.b = predicted bounding box****

YOLO: Generating Bounding Boxes

$$Y = [pc, bx, by, bh, bw, c1]$$

Probability the
box contains
an object

X coordinate
of p.b.b

Y coordinate
of p.b.b.

**Height of
p.b.b.**

****p.b.b = predicted bounding box****

YOLO: Generating Bounding Boxes

$$Y = [pc, bx, by, bh, bw, c1]$$

Probability the
box contains
an object

X coordinate
of p.b.b.

Y coordinate
of p.b.b.

Height of
p.b.b.

**Width of
p.b.b.**

****p.b.b = predicted bounding box****

YOLO: Generating Bounding Boxes

$$Y = [pc, bx, by, bh, bw, c1]$$

Probability the
box contains
an object

X coordinate
of p.b.b.

Y coordinate
of p.b.b.

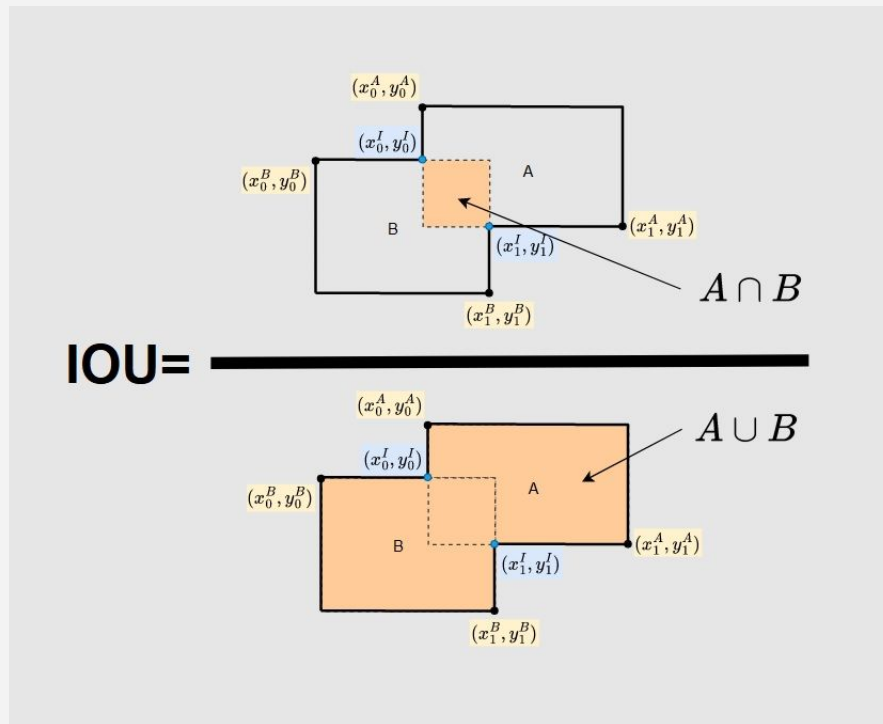
Height of
p.b.b.

Width of
p.b.b.

**Class 1 label:
"Fractured"**

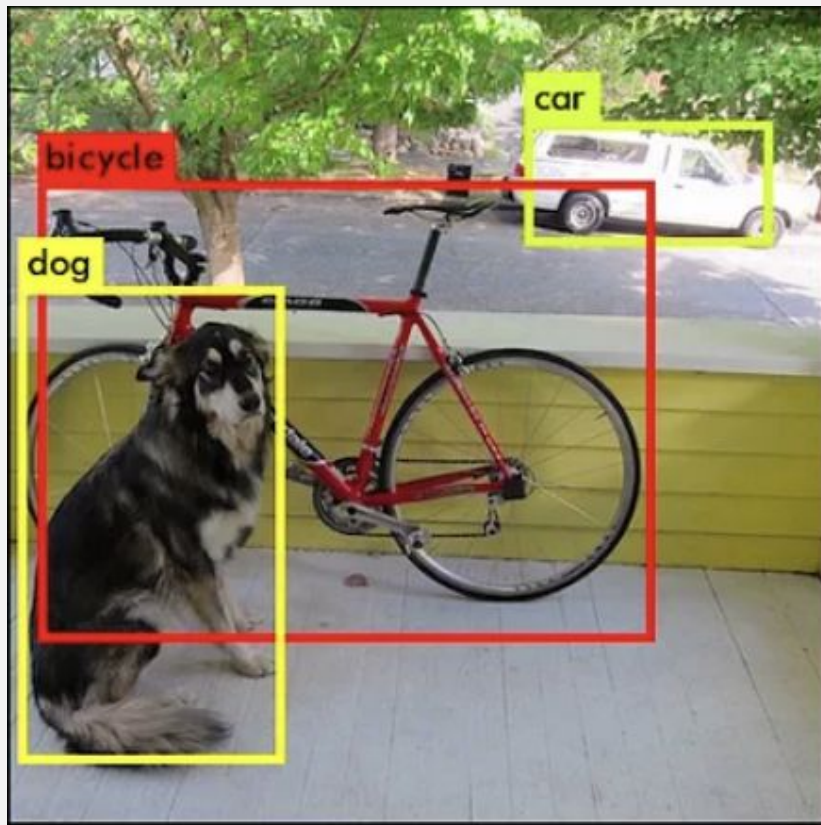
****p.b.b = predicted bounding box****

YOLO: Non-Max Suppression





YOLO: Select Best Box





04

DATASET

The Evaluation of Bone Fracture Detection of YOLO Series

Pongsakorn Samothai
Panyapiwat Institute of Management
Nontaburi, Thailand
6252300462@stu.pim.ac.th

Kanabadee Srisomboon
King Mongkut's University of
Technology North Bangkok
Bangkok, Thailand
kanabadee.s@eng.kmutnb.ac.th

Parinya Sanguansat
Panyapiwat Institute of Management
Nontaburi, Thailand
parinyasan@pim.ac.th

Wilaiporn Lee
King Mongkut's University of
Technology North Bangkok
Bangkok, Thailand
wilaiporn.l@eng.kmutnb.ac.th

Adisorn Kheaksong
Panyapiwat Institute of Management
Nontaburi, Thailand
adisornkhe@pim.ac.th

Abstract— Convolutional Neural Network (CNN) becomes the most powerful tool for object detection application and is mostly used in several filed such as security and medical. Bone fracture detection is one of the most medical application which will be useful in the medical treatment since there are number of misdetections according to human errors and small fracture locations. YOLO is the advanced CNN model that is interesting for researchers to be implemented with several applications. The YOLO-X and YOLO-R are the current models which are implemented in many applications. However, these models have not been evaluated on the low feature images such as bone X-ray images. Therefore, in this paper, we comprehensively study the performance of YOLO-X and YOLO-R under fracture bone X-ray images to locate the fracture locations. As a result, YOLO-X presents faster convergence time with much higher accuracy than YOLO-R since the YOLO-X model locates the fracture area with alternative processing such as detection head decoupling, anchors-free and augmentation strategies. Then, it can detect the fracture locations even if the features of X-ray images are low.

Keywords— Machine learning, CNN, YOLO, bone fracture.

I. INTRODUCTION

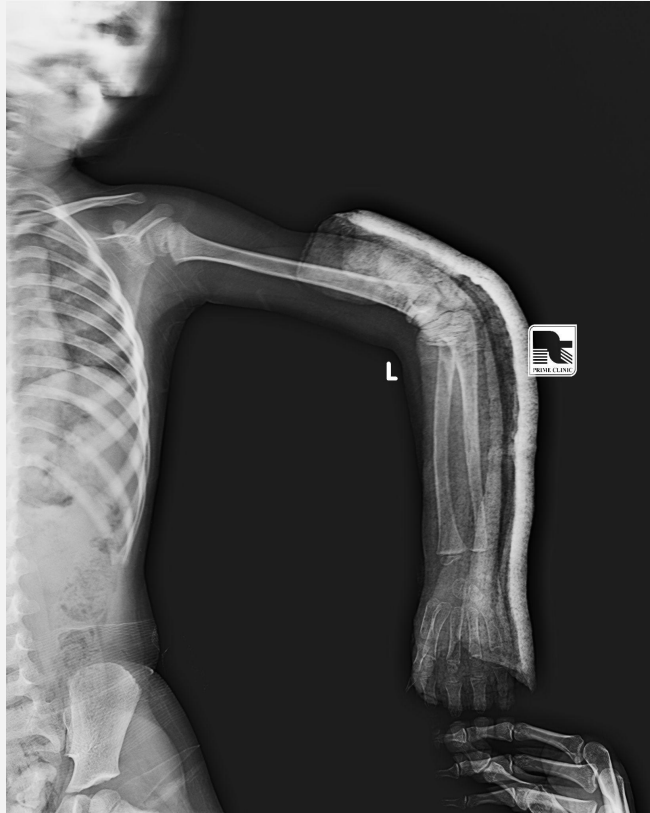
application with machine learning is exploited, not only the misdiagnosis can be reduced, the diagnosis time also significantly reduces. Once the X-ray image is considered as a grayscale image with low features, it is very difficult to classify the fracture area with a conventional CNNs.

A few early studies have shown the feasibility of CNNs usage to detect fractures on radiographic images. In [5], Y. L. Thian and team presented an automated fracture detection and localization on wrist radiographs using model Inception-ResNet Faster R-CNN which were able to attain the area under the receiver operating characteristic curve (AUC) of 0.91. Rebecca and Anuj [1] presented an assessment of a deep-learning system for fracture detection in musculoskeletal radiographs which were able to attain an area under the receiver operating characteristic curve (AUC) of 0.974.

Yolo models [6] are the most popular machine learning model exploiting the CNNs to detect objects such as car, licensed plate, human, face and so on. Different series of Yolo model are presented with alternative solutions to improve the performance of the traditional CNN. Lately, YOLO-R [7] and YOLO-X [8] are the two most interesting techniques among other series to perform the object

Collected Data ...

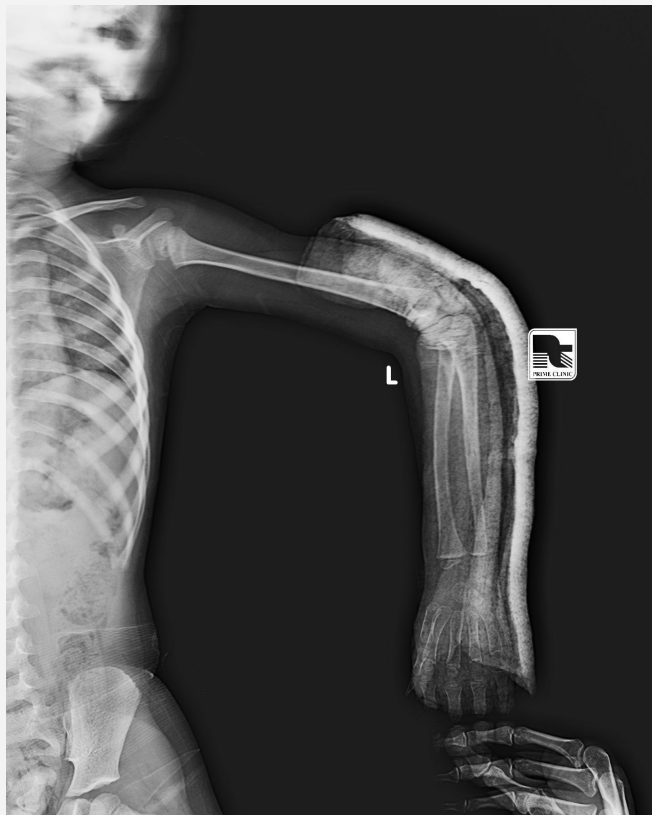




FRACTURED



NOT FRACTURED



FRACTURED



NOT FRACTURED



train
image_id
IMG0000019.jpg
IMG0000025.jpg
IMG0000044.jpg
IMG0000057.jpg
IMG0000058.jpg
IMG0000059.jpg
IMG0000092.jpg
IMG0000100.jpg

valid
image_id
IMG0003733.jpg
IMG0003734.jpg
IMG0003766.jpg
IMG0003767.jpg
IMG0003770.jpg
IMG0003771.jpg
IMG0003774.jpg
IMG0003785.jpg

THE SPLIT



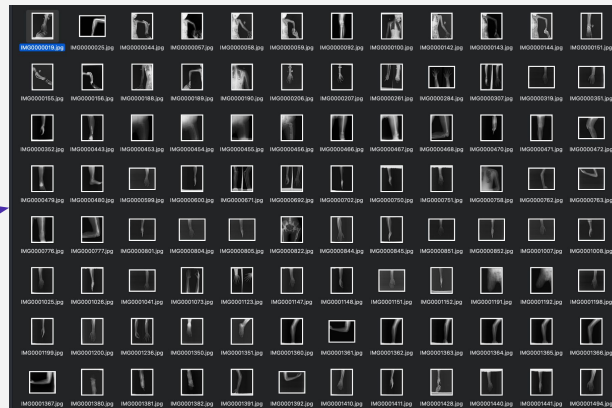
train
image_id
IMG0000019.jpg
IMG0000025.jpg
IMG0000044.jpg
IMG0000057.jpg
IMG0000058.jpg
IMG0000059.jpg
IMG0000092.jpg
IMG0000100.jpg

valid
image_id
IMG0003733.jpg
IMG0003734.jpg
IMG0003766.jpg
IMG0003767.jpg
IMG0003770.jpg
IMG0003771.jpg
IMG0003774.jpg
IMG0003785.jpg



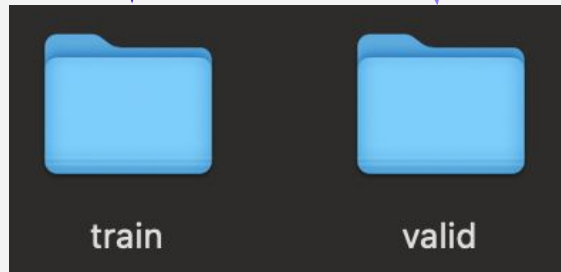
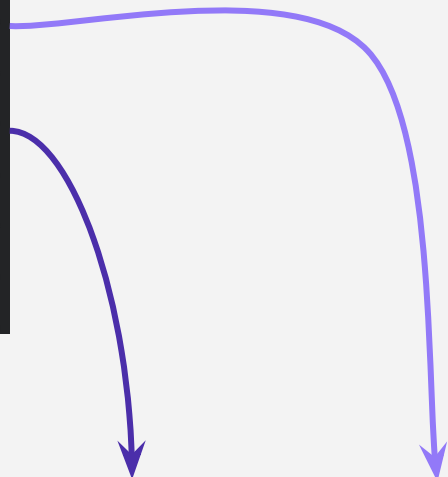
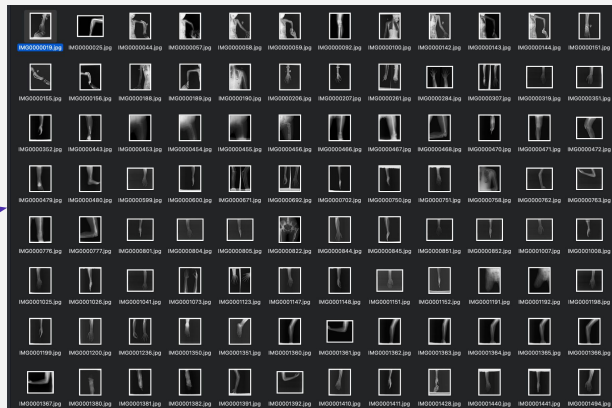
train	
image_id	
IMG0000019.jpg	
IMG0000025.jpg	
IMG0000044.jpg	
IMG0000057.jpg	
IMG0000058.jpg	
IMG0000059.jpg	
IMG0000092.jpg	
IMG0000100.jpg	

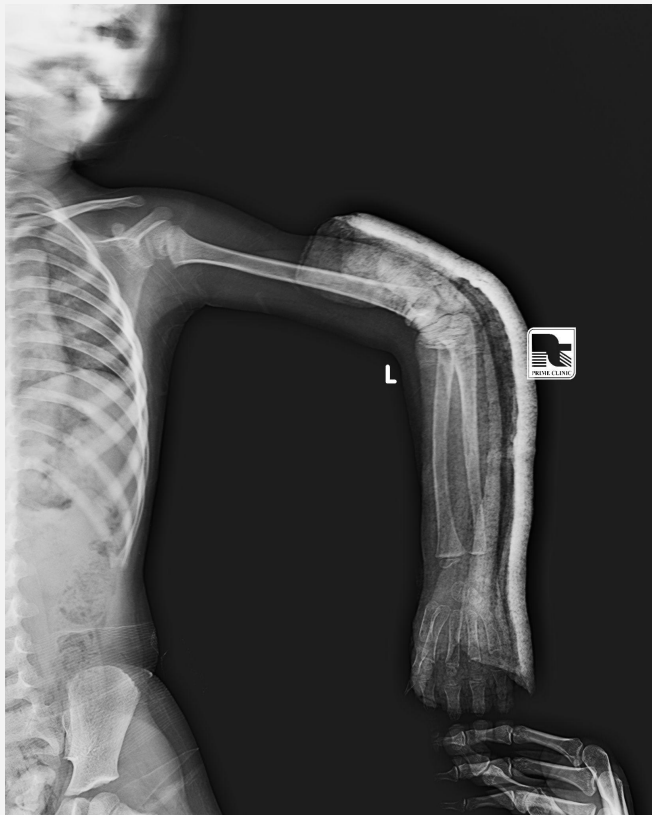
valid	
image_id	
IMG0003733.jpg	
IMG0003734.jpg	
IMG0003766.jpg	
IMG0003767.jpg	
IMG0003770.jpg	
IMG0003771.jpg	
IMG0003774.jpg	
IMG0003785.jpg	



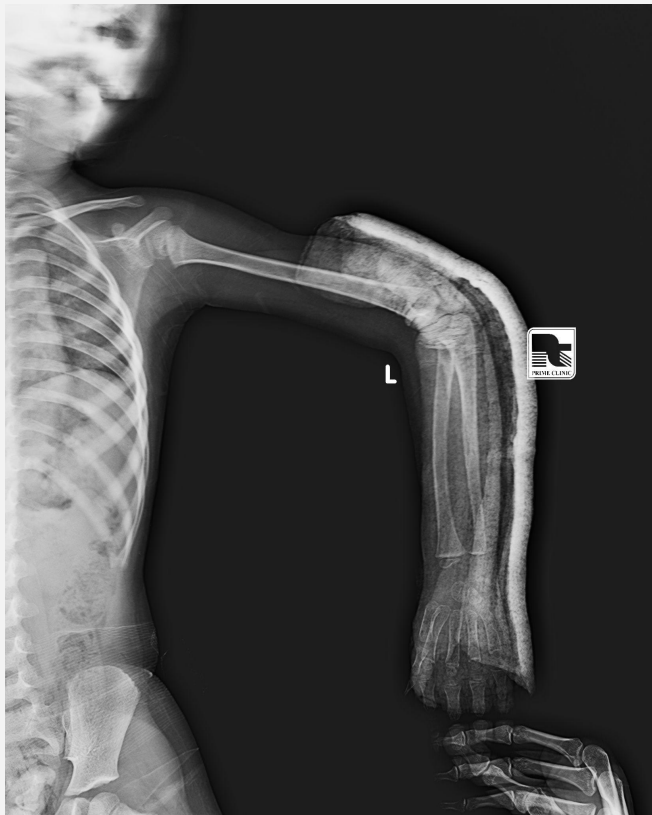
train
image_id
IMG0000019.jpg
IMG0000025.jpg
IMG0000044.jpg
IMG0000057.jpg
IMG0000058.jpg
IMG0000059.jpg
IMG0000092.jpg
IMG0000100.jpg

valid
image_id
IMG0003733.jpg
IMG0003734.jpg
IMG0003766.jpg
IMG0003767.jpg
IMG0003770.jpg
IMG0003771.jpg
IMG0003774.jpg
IMG0003785.jpg





**NOW WE HAVE
X_TRAIN AND X_VALID**



NOW WE HAVE
X_TRAIN AND X_VALID

WHAT'S OUR Y?

```
{
  "info": { "description": "my-project-name" },
  "images": [
    { "id": 1, "width": 2304, "height": 2880, "file_name": "IMG0000019.jpg" },
    { "id": 2, "width": 2880, "height": 2304, "file_name": "IMG0000025.jpg" },
    { "id": 3, "width": 2304, "height": 2880, "file_name": "IMG0000044.jpg" },
    { "id": 4, "width": 2304, "height": 2880, "file_name": "IMG0000057.jpg" },
    { "id": 5, "width": 2304, "height": 2880, "file_name": "IMG0000058.jpg" },
    { "id": 6, "width": 2304, "height": 2880, "file_name": "IMG0000059.jpg" },
  ]
}
```

CREATING THE Y VARIABLE

```
{  
  "info": { "description": "my-project-name" },  
  "images": [  
    { "id": 1, "width": 2304, "height": 2880, "file_name": "IMG0000019.jpg" },  
    { "id": 2, "width": 2880, "height": 2304, "file_name": "IMG0000025.jpg" },  
    { "id": 3, "width": 2304, "height": 2880, "file_name": "IMG0000044.jpg" },  
    { "id": 4, "width": 2304, "height": 2880, "file_name": "IMG0000057.jpg" },  
    { "id": 5, "width": 2304, "height": 2880, "file_name": "IMG0000058.jpg" },  
    { "id": 6, "width": 2304, "height": 2880, "file_name": "IMG0000059.jpg" },  
  ]  
}
```



convert_coco_to_yolo()

**CREATING THE
Y VARIABLE**





```
{
  "info": { "description": "my-project-name" },
  "images": [
    { "id": 1, "width": 2304, "height": 2880, "file_name": "IMG0000019.jpg" },
    { "id": 2, "width": 2880, "height": 2304, "file_name": "IMG0000025.jpg" },
    { "id": 3, "width": 2304, "height": 2880, "file_name": "IMG0000044.jpg" },
    { "id": 4, "width": 2304, "height": 2880, "file_name": "IMG0000057.jpg" },
    { "id": 5, "width": 2304, "height": 2880, "file_name": "IMG0000058.jpg" },
    { "id": 6, "width": 2304, "height": 2880, "file_name": "IMG0000059.jpg" },
  ]
}
```

CREATING THE Y VARIABLE

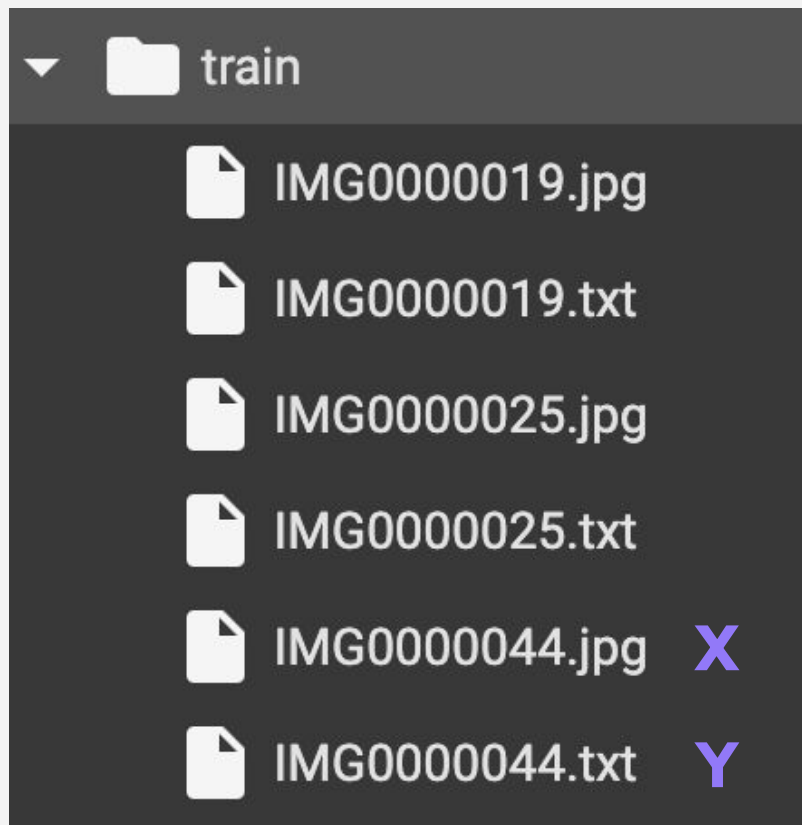


convert_coco_to_yolo()

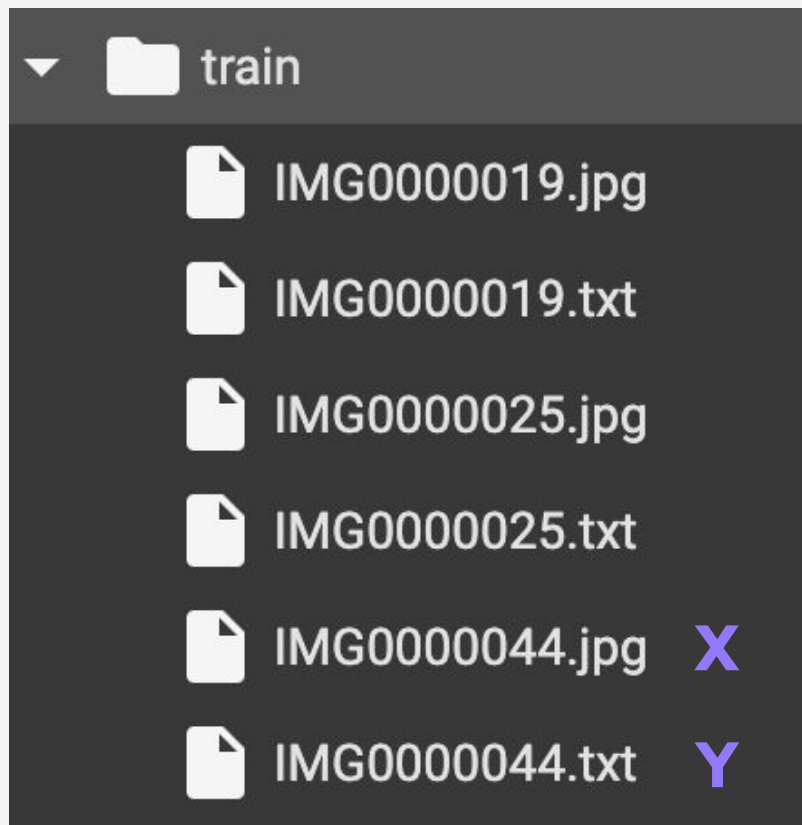


```
0 0.5985 0.3482 0.1186 0.0512
```





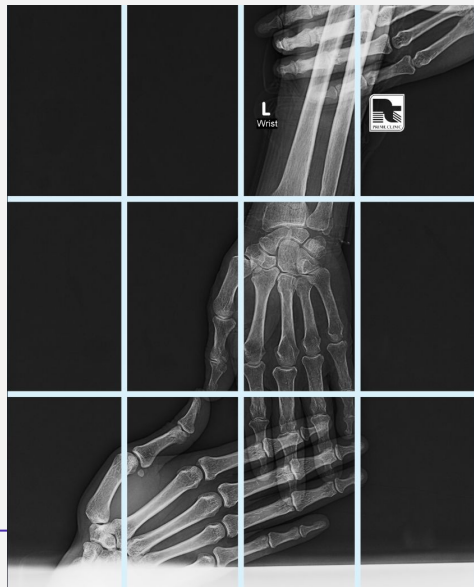
**COMBINE
X_TRAIN &
Y_TRAIN**



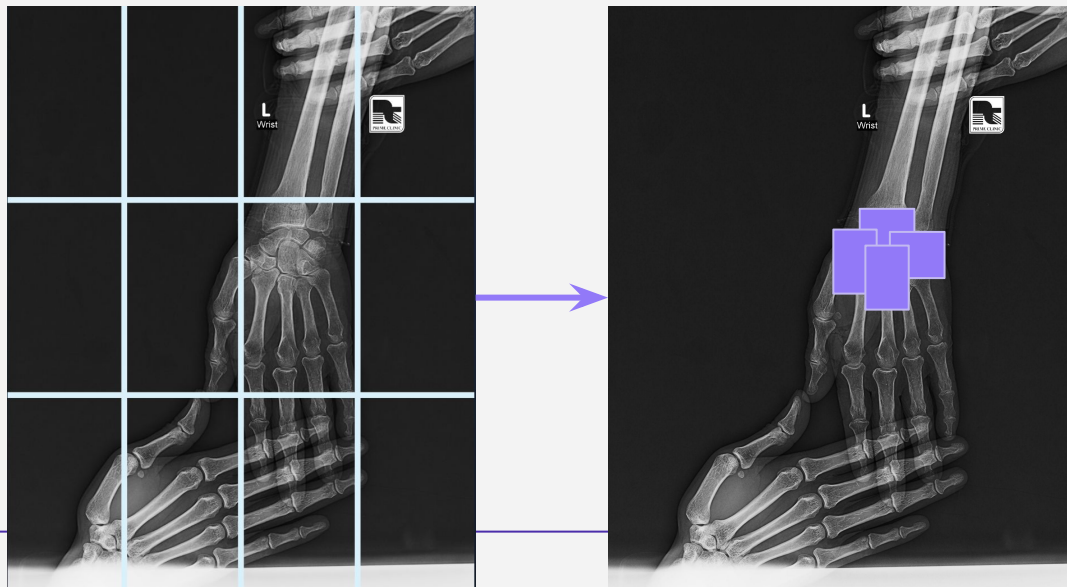
**COMBINE
X_TRAIN &
Y_TRAIN**

**SPECIFY PATHS
IN .YAML FILE**

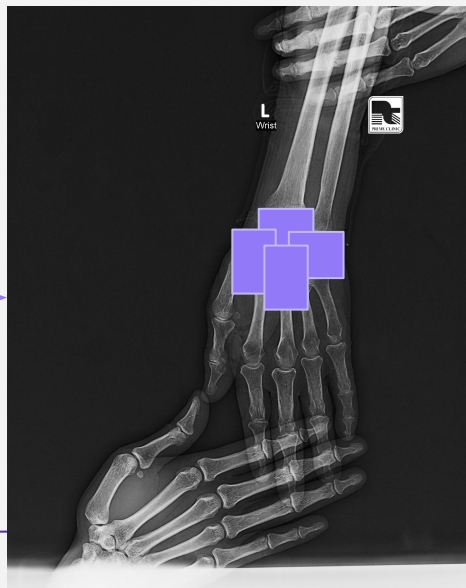
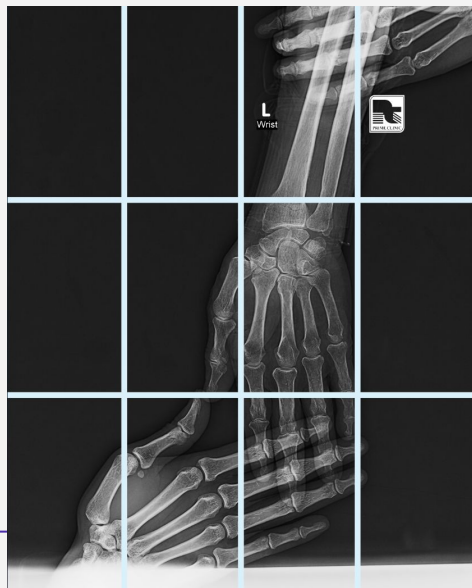
YOLO: Applied to Our Data



YOLO: Applied to Our Data



YOLO: Applied to Our Data





05

RESULTS

FINAL HYPERPARAMETERS

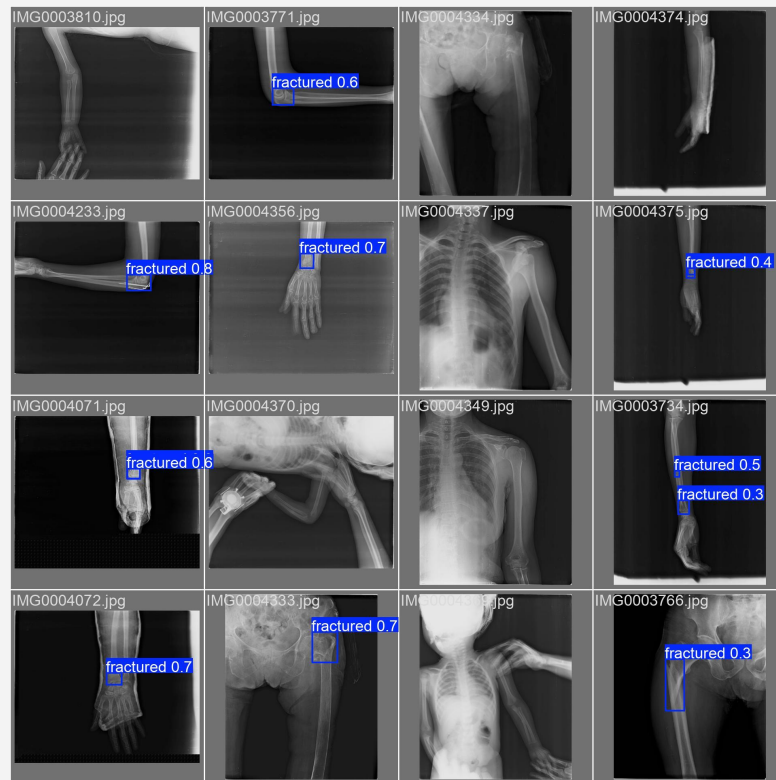
Learn Rate = 0.002

Optimizer = AdamW

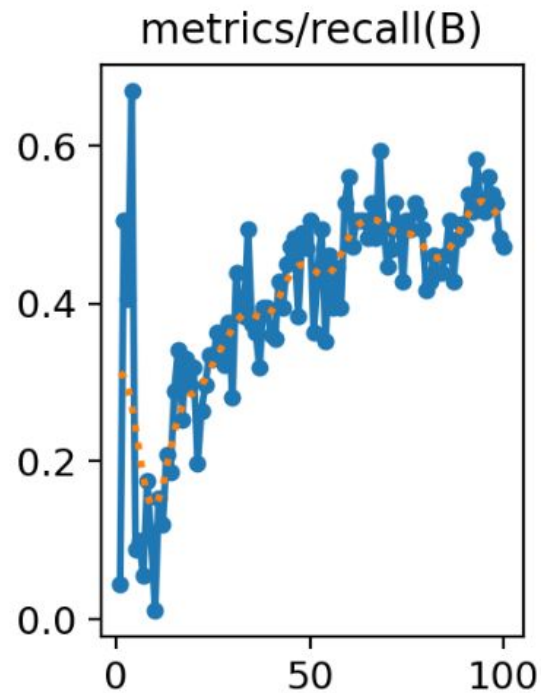
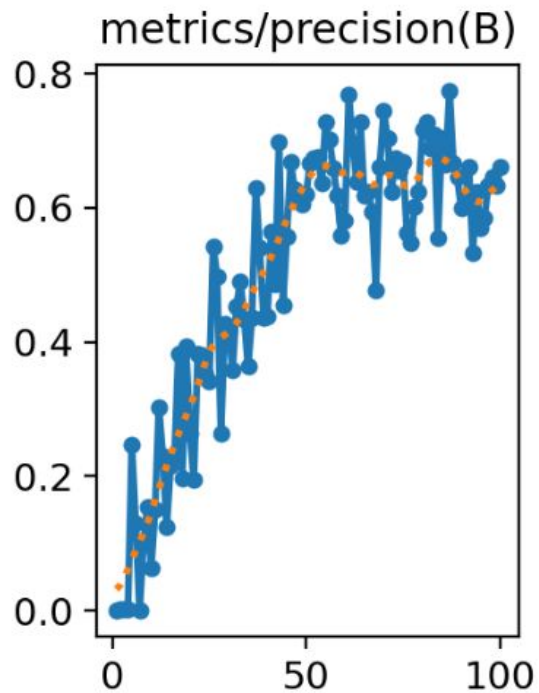
Momentum = 0.9

Weight decay = 0

OUR RESULTS



OUR RESULTS

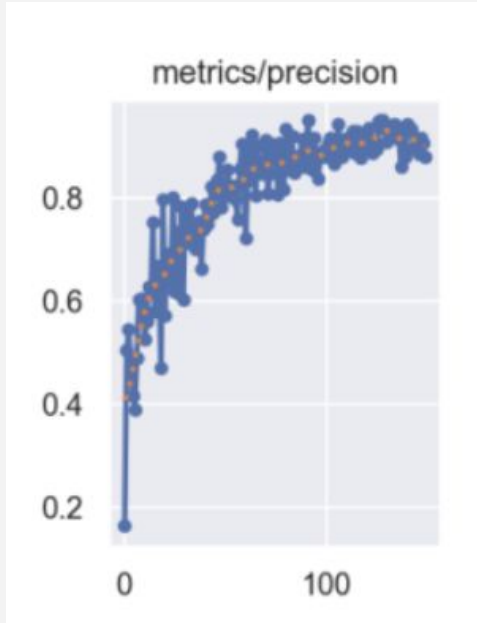




06

COMPARISON

LITERATURE REVIEW



Liu, 2024

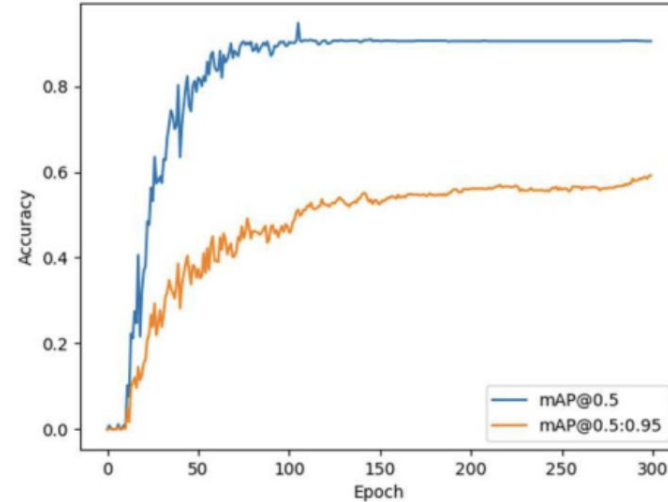


Fig. 3. YOLO-X training accuracy graph

Samatai et al, 2022

CITATIONS



Dey, Ishani. "What is YOLO Algorithm." Medium, Jul 12, 2023. <https://medium.com/@ishudey11032002/what-is-yolo-algorithm-ef5a3326510b>

Liu, Bo. (2024). DW-Yolo: Improved yolo for bone fracture detection. Sixth International Conference on Information Science, Electrical, and Automation Engineering (ISEAE 2024), 23. <https://doi.org/10.1117/12.3037465>

Parti, Ayush. "Introduction to YOLO Object Detection: Understanding the Basics." Pareto, May 8, 2024. <https://pareto.ai/blog/yolo-object-detection>

Samothai, P., Sanguansat, P., Kheaksong, A., Srisomboon, K., & Lee, W. (2022). The evaluation of bone fracture detection of Yolo Series. 2022 37th International Technical Conference on Circuits/Systems, Computers and Communications (ITC-CSCC). <https://doi.org/10.1109/itc-cscc55581.2022.9895016>

<https://pmc.ncbi.nlm.nih.gov/articles/PMC10178524/#:~:text=In%20medical%20imaging%2C%20YOLO%20has,eye%20images%3B%20Bonrod%20et%20al.>

<https://builtin.com/machine-learning/non-maximum-suppression>

<https://kili-technology.com/data-labeling/machine-learning/yolo-algorithm-real-time-object-detection-from-a-to-z>

<https://www.youtube.com/watch?v=ag3DLKsl2vk>





THANK YOU!

